Chapter 11. Smoothing and Density Estimation

Histograms, kernel density estimation, bandwidth selection, multivariate KDE, confidence bands

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11.1 From histograms to kernels

The EDF has no derivative, and a histogram depends on bin width and origin; averaging histograms over shifted origins gives the kernel estimator.

Definition (Kernel density estimator)

For a kernel K and bandwidth $h > 0$, $\hat{f}(x) = \frac{1}{nh}\sum_{i=1}^{n} K\Big(\frac{x - X_i}{h}\Big)$.

Standard kernels: $K \geq 0$, $\int K = 1$, $K(-u) = K(u)$, $\sigma_K^2 = \int u^2 K < \infty$, e.g. Gaussian ϕ, Epanechnikov $\frac{3}{4}(1-u^2)_+$, uniform $\frac{1}{2}I(|u| \leq 1)$. Then $\hat{f}$ is a density. The choice of h matters far more than that of K.

Example 11.1

Data $\{0, 1, 3\}$, Gaussian kernel, $h = 1$: $\hat{f}(1) = \frac{1}{3}[\phi(1)+\phi(0)+\phi(2)] = \frac{1}{3}(0.242+0.399+0.054) = 0.232$.

11.2 Bias, variance and bandwidth

Motivation. Bias and variance determine the optimal bandwidth and the $n^{-4/5}$ rate.

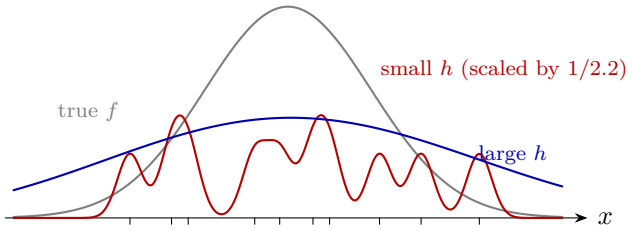


Figure. *Kernel estimates from ten points (ticks): small h is spiky (high variance; vertically shrunk), large h flattens the peak (bias).*

Theorem 11.1

Let f be twice continuously differentiable near x with $f(x) > 0$. As $n \to \infty$, $h \to 0$ and $nh \to \infty$,

$$\operatorname{bias} \hat{f}(x) = \frac{\sigma_K^2 f''(x)}{2} h^2 + o(h^2), \qquad \operatorname{Var} \hat{f}(x) = \frac{f(x) R(K)}{nh} + o\Big(\frac{1}{nh}\Big), \qquad R(K) = \int K^2.$$

Proof [Supplement]. $E\hat{f}(x) = \int K(u) f(x - hu)\, du$; expand

$$f(x - hu) = f(x) - huf'(x) + \tfrac{h^2u^2}{2} f''(x) + o(h^2u^2),$$

use $\int uK = 0$, $\int u^2 K = \sigma_K^2$ and dominated convergence. $\operatorname{Var} \hat{f}(x) = \frac{1}{n} \operatorname{Var}\big(\frac{1}{h} K(\frac{x-X}{h})\big)$, with $E\frac{1}{h^2} K^2(\frac{x-X}{h}) = \frac{1}{h}\int K^2(u) f(x - hu)\, du = \frac{f(x)R(K)}{h} + O(1)$ and squared mean $O(1)$. ∎

This is the **bias–variance trade-off**. Integrating,

$$\operatorname{MISE}(\hat{f}) = E\int (\hat{f} - f)^2 \approx \frac{\sigma_K^4 R(f'')}{4} h^4 + \frac{R(K)}{nh}, \qquad R(f'') = \int (f'')^2,$$

minimized at

$$h^* = \Big(\frac{R(K)}{\sigma_K^4 R(f'')\,n}\Big)^{1/5} \asymp n^{-1/5}, \qquad \mathrm{MISE}(h^*) \asymp n^{-4/5}. \tag{11.1}$$

(Set $\sigma_K^4 R(f'')h^3 - R(K)/(nh^2) = 0$.) The rate $n^{-4/5}$ is slower than the parametric n^{-1}.

Choosing h. h^* contains the unknown $R(f'')$.

- **Normal reference.** $f = N(\mu, \sigma^2)$ gives $R(f'') = \frac{3}{8\sqrt{\pi}\sigma^5}$, so with the Gaussian kernel $h = 1.06\,\hat\sigma n^{-1/5}$, $\hat\sigma = \min(s, \mathrm{IQR}/1.34)$.
- **Silverman's rule:** $h = 0.9\,\hat\sigma n^{-1/5}$, protecting against multimodality.
- **Plug-in (Sheather–Jones):** estimate $R(f'')$ by a pilot estimate of f'' in (11.1); generally recommended. Cross-validation is an alternative.

Proof of the normal value of $R(f'')$. For $\sigma = 1$, $\phi''(x) = (x^2-1)\phi(x)$ and $\phi^2 = \frac{1}{2\sqrt{\pi}}\times$ the $N(0, \frac12)$ density, so $R(\phi'') = \frac{1}{2\sqrt{\pi}} E(Z^2-1)^2 = \frac{1}{2\sqrt{\pi}}(\frac34 - 1 + 1) = \frac{3}{8\sqrt{\pi}}$, $Z \sim N(0, \frac12)$; scaling gives σ^{-5}. With $R(\phi) = \frac{1}{2\sqrt{\pi}}$, $\sigma_K = 1$, $h^* = (4/3)^{1/5}\sigma n^{-1/5} = 1.06\sigma n^{-1/5}$. ∎

11.3 Multivariate density estimation

For $\mathbf{X}_i \in \mathbb{R}^p$, $\hat f(\mathbf{x}) = \frac1n \sum_i K_H(\mathbf{x} - \mathbf{X}_i)$, $K_H(\mathbf{x}) = |H|^{-1/2} K(H^{-1/2}\mathbf{x})$, with bandwidth matrix H (diagonal, or a full plug-in matrix). With a product kernel and equal widths h,

$$\mathrm{MSE}\,\hat f(\mathbf{x}) \approx \Big[\frac{\sigma_K^2 h^2}{2}\sum_j \partial_{jj} f(\mathbf{x})\Big]^2 + \frac{f(\mathbf{x})R(K)^p}{nh^p},$$

so $h^* \asymp n^{-1/(p+4)}$, $\mathrm{MSE} \asymp n^{-4/(p+4)}$: the **curse of dimensionality** (for $p = 10$ a cube window of side 0.1 holds $10^{-10}n$ points on average). Practical for $p \le 2$, occasionally 3.

11.4 Confidence intervals and bands

At h^* bias and SE have the same order, so intervals target $f_h(x) = E\hat f(x)$.

- **Pointwise, plug-in:** by the CLT, $\sqrt{nh}(\hat f(x) - f_h(x)) \xrightarrow{d} N(0, f(x)R(K))$, so $f_h(x) \in \hat f(x) \pm z_{\alpha/2}\sqrt{\hat f(x)R(K)/(nh)}$.
- **Pointwise, bootstrap:** recompute $\hat f^*$ on a grid with the same h; use $\hat f \pm z_{\alpha/2}\widehat{\mathrm{se}}^*$, the basic $(2\hat f - Q^*_{1-\alpha/2}, 2\hat f - Q^*_{\alpha/2})$, or $\hat f \pm Q_{1-\alpha}(|\hat f^* - \hat f|)$.
- **Simultaneous band:** with q^* the $1-\alpha$ quantile of $\Delta^* = \max_{x\in\mathrm{grid}}|\hat f^*(x) - \hat f(x)|$, $\hat f(x) \pm q^*$ covers f_h on the grid with asymptotic probability $1-\alpha$; much wider.

Negative lower limits can be truncated at 0.

A bootstrap-t with $t^* = (\hat f^* - \hat f)/\sqrt{\hat f^* R(K)/(nh)}$ is unstable: where $\hat f^*(x) \approx 0$, t^* explodes; this also rules out studentized-maximum bands. Pointwise intervals on a grid are reported more often (some authors call them a "confidence band").

11.5 Selected exercises

Motivation. The $n^{-4/5}$ rate, why cross-validation targets MISE, and how little the kernel matters.

Exercise 11.5 (The $n^{-4/5}$ rate)

(a) Show that the asymptotic MISE at h^* in (11.1) equals $\frac{5}{4}\{\sigma_K^4 R(f'')\}^{1/5} R(K)^{4/5} n^{-4/5}$.
(b) For x with $f(x) > 0$ and f as in Theorem 11.1, show that MSE $\hat{f}(x) = O(n^{-4/5})$ for any $h = Cn^{-1/5}$.
(c) Compare with $\mathrm{MSE}(\bar{X}_n) = \sigma^2/n$.

Solution.

(a) Let $A(h) = ah^4 + c/(nh)$, $a = \sigma_K^4 R(f'')/4$, $c = R(K)$. At the minimizer $4ah^{*3} = c/(nh^{*2})$, i.e. $ah^{*4} = c/(4nh^*)$, so

$$A(h^*) = \frac{c}{4nh^*} + \frac{c}{nh^*} = \frac{5}{4} \cdot \frac{c}{nh^*}.$$

With $h^* = \{c/(4an)\}^{1/5}$, $A(h^*) = \frac{5}{4}(4a)^{1/5} c^{4/5} n^{-4/5}$, $4a = \sigma_K^4 R(f'')$.

(b) By Theorem 11.1,

$$\mathrm{MSE}\, \hat{f}(x) = \mathrm{bias}^2 + \mathrm{Var} = \Big(\frac{\sigma_K^2 f''(x)}{2}\Big)^2 h^4 + \frac{f(x) R(K)}{nh} + o\Big(h^4 + \frac{1}{nh}\Big).$$

With $h = Cn^{-1/5}$, $h^4 = C^4 n^{-4/5}$ and $1/(nh) = C^{-1} n^{-4/5}$.

(c) $\mathrm{MSE}\, \hat{f}(x) / \mathrm{MSE}(\bar{X}_n) \asymp n^{1/5} \to \infty$: halving the MSE needs $2^{5/4} \approx 2.38$ times the data, against 2 for a mean. ■

Exercise 11.7 (Least squares cross-validation)

Let $\hat{f}_{-i}(x) = \frac{1}{(n-1)h} \sum_{j \neq i} K\big(\frac{x - X_j}{h}\big)$ and

$$\mathrm{CV}(h) = \int \hat{f}^2 - \frac{2}{n} \sum_{i=1}^{n} \hat{f}_{-i}(X_i).$$

Assuming K is bounded and $\int f^2 < \infty$, show that $E\,\mathrm{CV}(h) = \mathrm{MISE}(h) - \int f^2$ for every h. Explain why the leave-one-out estimate is needed.

Solution [Supplement].

Step 1. $\mathrm{MISE}(h) = E\int \hat{f}^2 - 2E\int \hat{f} f + \int f^2$, and with $K_h(u) = K(u/h)/h$ and Fubini,

$$E\int \hat{f}(x) f(x)\, dx = \int E\hat{f}(x)\, f(x)\, dx = \iint K_h(x - t) f(t) f(x)\, dt\, dx.$$

Step 2. For $j \neq i$, $EK_h(X_i - X_j) = \iint K_h(x - t) f(x) f(t)\, dx\, dt$, so $E\hat{f}_{-i}(X_i)$ is the same integral and $E\big[\frac{2}{n}\sum_i \hat{f}_{-i}(X_i)\big] = 2E\int \hat{f} f$.

Step 3. So $E\,\mathrm{CV}(h) = \mathrm{MISE}(h) - \int f^2$, the last term free of h.

Using $\hat{f}(X_i) = \frac{K(0)}{nh} + \frac{n-1}{n}\hat{f}_{-i}(X_i)$ would subtract an extra $\frac{2K(0)}{nh} \to \infty$ as $h \to 0$, driving $h \to 0$: each point "predicted" by its own spike. ■

Exercise 11.8 (How much does the kernel matter?)

(a) Show that $\sigma_K R(K)$ is unchanged by rescaling $K_c(u) = K(u/c)/c$, and that the optimal AMISE is proportional to $\{\sigma_K R(K)\}^{4/5}$.
(b) Compute $\sigma_K R(K)$ for the Gaussian, Epanechnikov and uniform kernels, and the relative sample sizes needed for equal optimal AMISE.
(c) Derive the normal-reference bandwidth for the Epanechnikov kernel.

Solution [Supplement].

(a) With $u = cv$, $\sigma_{K_c}^2 = c^2\sigma_K^2$ and $R(K_c) = R(K)/c$, so $\sigma_{K_c}R(K_c) = \sigma_K R(K)$: rescaling only changes the meaning of h. By Exercise 11.5(a) the optimal AMISE is

$$\frac{5}{4}R(f'')^{1/5}(\sigma_K^4)^{1/5}R(K)^{4/5}n^{-4/5} = \frac{5}{4}R(f'')^{1/5}\{\sigma_K R(K)\}^{4/5}n^{-4/5}.$$

(b) Gaussian: $\sigma_K = 1$, $R(K) = \frac{1}{2\sqrt{\pi}}$, product 0.2821. Epanechnikov: $\sigma_K^2 = \frac{3}{4}\int_{-1}^{1}(u^2 - u^4)\,du = \frac{1}{5}$, $R(K) = \frac{9}{16}\int_{-1}^{1}(1-u^2)^2\,du = \frac{3}{5}$, product $\frac{3}{5\sqrt{5}} = 0.2683$. Uniform: $\sigma_K^2 = \frac{1}{3}$, $R(K) = \frac{1}{2}$, product 0.2887. Equal AMISE needs $n \propto \sigma_K R(K)$: relative to Epanechnikov, Gaussian 1.05, uniform 1.08. The kernel costs at most a few percent.

(c) With $R(f'') = \frac{3}{8\sqrt{\pi}\sigma^5}$ (Section 11.2), (11.1) gives

$$h^* = \Big(\frac{3/5}{(1/25)\cdot 3/(8\sqrt{\pi}\sigma^5)\cdot n}\Big)^{1/5} = (40\sqrt{\pi})^{1/5}\sigma n^{-1/5} = 2.34\,\sigma n^{-1/5}.$$

It exceeds the Gaussian $1.06\,\sigma n^{-1/5}$ only because of scale ($\sigma_K = 0.447$ vs. 1), as (a) explains. ■